Md. Mutasim Billah Abu Noman Akanda

PhD Applicant (Fall 2027) • AI/ML Research

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Research Interests

Document AI and OCR for low-resource languages, robust ML under distribution shift, and efficient multimodal systems (LLM/VLM) with reproducible evaluation. Current focus: document layout analysis, handwritten OCR, retrieval-augmented systems, and practical methods that connect scientific rigor to real-world deployment constraints.

Education

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
  B.Sc. Computer Science and Engineering *Jun 2019 – Feb 2023*

  CGPA: 3.96/4.00 **Distinction:** Summa Cum Laude (top 1%; conferred May 2024)
  Honors: Vice Chancellor's Honors List; Dean's List Scholarship (3×)
  Relevant coursework: Machine Learning, Artificial Intelligence, Computer Vision, Natural Language Processing, Algorithms, Software Engineering

Publications

- **Akanda, M.B.A.N., Ahmed, M., Rabby, A.S.A., & Rahman, F.**: *Optimum Deep Learning Method for Document Layout Analysis in Low Resource Languages.* ACM Southeast Conference (ACM SE'24), 199–204. doi:10.1145/3603287.3651184
- **Akanda, M.B.A.N., Prodhan, M., Sarwar, S., Raatul, A.M., Paul, B.**: *Voice Controlled Home Automation with Cloud-Based Environment Monitoring System.* ICTCS 2022, Lecture Notes in Networks and Systems, vol. 623, Springer, Singapore. doi:10.1007/978-981-19-9638-2_21

Research Profile

- **Primary theme**: Document intelligence in low-resource settings: layout analysis, printed/handwritten OCR, and robust extraction from noisy real-world documents.
- **Secondary theme**: Efficient multimodal ML systems (LLM/VLM and speech pipelines), including quantization, retrieval augmentation, and reliability-aware evaluation.
- **Research style**: Problem-driven experimentation with emphasis on reproducibility, baseline comparisons, and deployment-aware validation.

Research and Professional Experience

- **Apurba Technologies Ltd.** Dhaka, Bangladesh
  Senior Machine Learning Engineer *Sep 2025 – present*
  - **Document AI and OCR systems**: Worked on enterprise Bengali OCR and document layout analysis pipelines at operational scale (**10,000+** documents/day), with focus on robust extraction from heterogeneous document formats.
  - **Efficiency-aware model optimization**: Applied quantization to multimodal vision-language components (Qwen2.5-VL), reducing memory footprint by approximately **4**× while preserving practical task performance.
  - **Evaluation-oriented engineering**: Developed alignment components that improved conversion reliability (**+25%** conversion accuracy in internal benchmarks), reducing downstream manual correction load.
  - **Research relevance**: Translated production bottlenecks (layout variance, OCR noise, memory limits) into experimentable questions around model robustness and efficiency.
- **Apurba Technologies Ltd.** Dhaka, Bangladesh
  Machine Learning Engineer *Mar 2023 – May 2024*
  - **Low-resource document layout analysis**: Built and evaluated YOLOv8-based layout models for Bengali document structures (**85%+** internal accuracy), supporting OCR quality improvements in deployment.
  - **Inference optimization**: Implemented 8-bit quantization and measured **2**× faster inference with reduced memory usage, enabling broader operational deployment.
  - **Research dissemination**: Contributed to studies that led to publications at ACM SE'24 and Springer (ICTCS/LNNS).

- **Pixalate Inc.** Remote (California, USA)
  Data Scientist *Mar 2025 – Aug 2025*
  - **Web-scale ML systems**: Designed AI-assisted ad-network detection workflows over large crawling pipelines (**2M+** sites), combining learned and deterministic methods.
  - **Methodological trade-off analysis**: Used hybrid detection strategies to balance precision, throughput, and infrastructure cost in production settings.
- **Green Pants Studio** Remote (Texas, USA)
  AI Engineer *Jun 2024 – May 2025*
  - **Applied ML for valuation and mapping**: Developed ML-supported product valuation and data mapping workflows for large eCommerce streams.
  - **Pipeline quality and reliability**: Improved scraper throughput and deployment reliability through modular services and containerized delivery.

Teaching Experience

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
  Teaching Assistant, Department of Computer Science and Engineering *Feb 2021 – Jan 2023*
  - **Courses supported**: Introduction to Programming, Object-Oriented Programming, Artificial Intelligence, Software Engineering
  - **Instructional duties**: Led laboratory sessions and tutorials for cohorts of **30+** students; prepared lab materials and evaluated assignments/projects.
  - **Mentoring**: Guided students on programming and introductory AI/ML projects during office hours and practical sessions.

Selected Projects (Research-Relevant)

- **SenseScan (Bengali handwritten OCR)**: Pipeline with EAST/LANMS detection, CRNN recognition, and reading-order assembly; designed for experimentation in low-resource OCR settings. GitHub
- **Voice Healthcare Agent**: Multimodal conversational pipeline (FastAPI + Next.js + local voice stack) useful for studying latency/reliability trade-offs in speech-enabled LLM systems. GitHub
- **GradConnectAI**: Evidence-based matching system for graduate supervision discovery; relevant to ranking/retrieval experimentation in academic recommendation settings. GitHub

Reproducibility and Tooling

- **Experiment support**: Use MLflow/Weights & Biases style tracking and test-driven workflows (pytest/Vitest) to improve repeatability and regression control in ML pipelines.
- **Deployment-aware reproducibility**: Maintain API-first project structures (FastAPI, Docker, CI) to align research prototypes with executable, verifiable systems.
- **Data and evaluation mindset**: Prioritize clear evaluation criteria, metric definitions, and error-analysis loops when moving from prototype to production conditions.

Technical Skills

- **Languages**: Python, TypeScript, SQL
- **ML/AI**: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, quantization workflows
- **LLM/NLP**: RAG pipelines, vector search (pgvector/ChromaDB), vLLM, Ollama
- **Systems**: FastAPI, Docker, PostgreSQL, Redis, SQLite, async crawling and data pipelines
- **Experimentation**: MLflow, Weights & Biases, pytest, Vitest

Honors and Awards

- **Summa Cum Laude**: Top 1% of graduating class (May 2024)
- **Vice Chancellor's Honors List**: Scholarship recipient
- **Dean's List Scholarship**: Awarded 3 times
- **Programming contests**: Problem setter and award recipient in ULAB Take Off contests